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Deep levels by proton and electron irradiation in 4H-SiC

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The effects on 4H-silicon carbide epilayers of irradiation with protons and electrons having particle energies, respectively, of 6.5 and 8.2 MeV were carefully studied and critically compared. In detail, the electronic levels associated with the irradiation-induced defects were analyzed by current-voltage characteristics and deep-level transient spectroscopy (DLTS) measurements up to 550 K. In the same temperature range the apparent free-carrier concentration was measured by capacitance-voltage characteristics in order to monitor compensation effects due to the deep levels associated with the induced defects. Introduction rate, enthalpy, and capture cross section of such deep levels were compared. We found that a set of deep levels (at $E_T=0.39$ eV, $E_T=0.65$ eV, and $E_T=0.75$ eV) is the same in both cases of proton and electron irradiations, whereas two other pairs of levels (S1, $E_T=0.20$ eV and S1*, $E_T=0.23$ eV; S5, $E_T=1.09$ eV and S5*, $E_T=0.89$ eV) appearing in the same temperature range within the DLTS spectra should be associated with different defect complexes according to the irradiation type. Some conclusions regarding the microscopic nature of the defects related to the deep levels have been drawn. © 2005 American Institute of Physics. [DOI: 10.1063/1.2014941]

I. INTRODUCTION

Silicon carbide (SiC) is of major importance to our age because of its characteristics of high electric-field strength and saturation drift velocity and very high radiation hardness,¹ which make it a useful material for radiation detectors as well as for devices working close to reactors and engines.² Application in many fields always requires the controlled introduction of “imperfections” in the crystal structure. For this reason the so-called *materials science of defects*³ is fundamental for device developments where the “defect” understanding is essential. Thus, the study of defects introduced in a controlled way is a powerful tool to relate the properties of the defects themselves to the device performance. As well known, particle irradiation is a straightforward tool to generate simple defects in a controlled way. For this reason it is widely used in silicon and recently also in silicon carbide⁴ in order to investigate the origin and, whenever possible, the structure of the defects. Our contribution deals with the deep energy levels associated with defects induced by irradiation with protons and electrons in 4H-SiC, the doping density of which is in the range of $n=(2-6)\times 10^{15}$ cm⁻³. Deep-level enthalpy, concentration, and apparent capture cross section have been monitored for both irradiation types by changing the particle fluence and put in relationship with the observed compensation effects. Some conclusions about the nature of the defects introduced have been drawn. The charge transport properties of the ma-

terial, a key point for device performance, have also been analyzed.

II. EXPERIMENT

The samples examined were 4H-SiC epilayers grown by chemical-vapor deposition (CVD) and provided by CREE Inc. The doping concentration of the epitaxial layer in the proton-irradiated diodes was $N_d=5\times 10^{15}$ cm⁻³. This layer, 7 μ m thick, was grown on a 400- μ m-thick 4H-SiC substrate with doping density $N_d=10^{19}$ cm⁻³. The declared micropipe density was 16–30 cm⁻². The Ti Schottky contact was 1000 Å thick, annealed at 400 °C in N₂ atmosphere with an area of 1 mm². Metal oxide field plates were synthesized in order to prevent the electric field crowding at the contact edges. The Ohmic contacts were obtained by growing a Ti/Ni/Ag multilayer (with thickness of 1000 Å/2000 Å/2 μ m, respectively). The diodes were irradiated at room temperature with 6.5-MeV protons at fluences ranging from $\Phi=10^{11}$ cm⁻² to $\Phi=3.2\times 10^{13}$ cm⁻² (Table I). From

TABLE I. Irradiation conditions.

Proton 6.5 MeV		Electron 8.2 MeV	
Sample	Fluence Φ (cm ⁻²)	Sample	Fluence (cm ⁻²)
A4	...	I4	...
A2	10^{11}	G7	4.75×10^{13}
A1	10^{12}	A4	2.38×10^{14}
A3	3.2×10^{13}	G4	9.50×10^{14}

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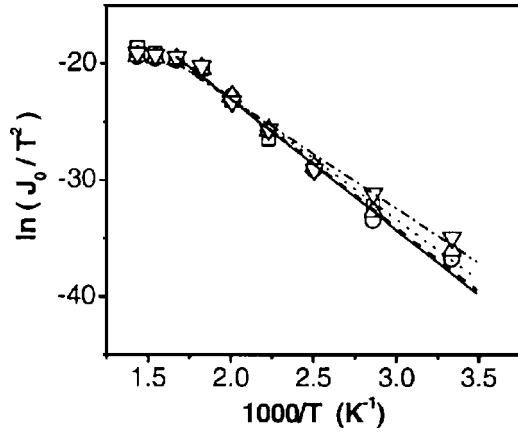


FIG. 1. Richardson's plot and its linear fit for the proton-irradiated diodes. Squares and continuous line: diode A1; circles and dashed line: diode A2; upwards triangles and dotted line: diode A3; and downwards triangles and dotted-dashed line: diode A4.

now on, we will refer straightforwardly to the irradiation fluences, for the sake of clarity. The electron-irradiated samples consisted of CVD-grown 30- μm -thick epitaxial layers with doping density $N_d = 2 \times 10^{15} \text{ cm}^{-3}$, a buffer layer with doping density $N_d = 10^{18} \text{ cm}^{-3}$, and a substrate with doping density $N_d = 10^{19} \text{ cm}^{-3}$. The substrate micropipe density was 10 cm^{-2} . The circular Au Schottky contact was 1500 Å thick with a diameter of 2 mm and one guard ring. The Ohmic contact was obtained by deposition of a 1000-Å-thick multilayer of Ti/Pt/Au, followed by annealing at 500 °C for 30 s in argon atmosphere. These samples were irradiated at 26 °C with electrons from a linear accelerator (LINAC) with energy equal to 8.2 MeV at fluences ranging from 4.75×10^{13} to $9.50 \times 10^{14} \text{ cm}^{-2}$ (Table I). The transport properties and the electronic levels associated with the defects were analyzed by current-voltage (I - V) characteristics and deep-level transient spectroscopy (DLTS) measurements up to 550 K, respectively. The deep-level enthalpy with respect to the conduction band E_T , the capture cross section σ , and the deep-level concentration N_T have been measured for each level. In the same temperature range the free-carrier concentration n was measured by capacitance-voltage

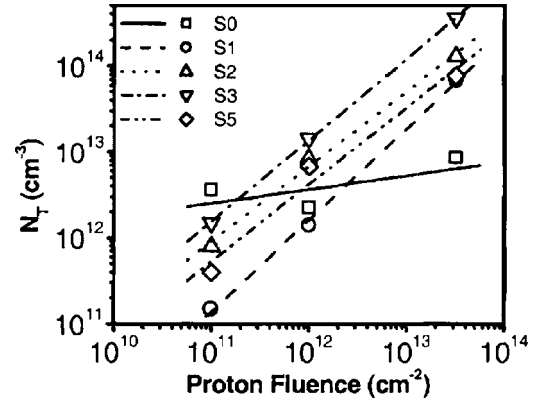


FIG. 2. Fluence dependence of the deep-level concentrations in 6.5-MeV irradiated samples. The lines are the linear fits to the experimental data.

(C - V) characteristics in the as-prepared and irradiated samples in order to monitor the compensation effects by intrinsic irradiation-induced defects. The DLTS measurements were carried out by means of a SULA double boxcar spectrometer with exponential correlator and the I - V characteristics were collected by a Keithley 6517 electrometer. The capacitance meter was a Keithley 3330, operating at a frequency $f = 100 \text{ kHz}$.

III. RESULTS AND DISCUSSION

Irradiation at high fluences expectedly introduces defects in the material, notwithstanding the high radiation hardness of 4H-SiC that manifests itself in the charge transport characteristics with respect to the irradiation dose. In Fig. 1 the Richardson plot of the temperature-corrected diode saturation current J_0/T^2 versus inverse temperature T^{-1} demonstrates that the activation energy for the charge transport remains almost constant within the range of 0.80–0.96 eV. The I - V characteristics did not show any significant difference between the as-prepared diode and the diode irradiated with the highest dose. The same behavior was detected in the electron-irradiated samples, whereby for fluences up to $9.5 \times 10^{14} \text{ cm}^{-2}$ no increase of the leakage current was observed for reverse bias in the range of 0–200 V.

TABLE II. Enthalpy E_T , concentration N_T , apparent capture cross section σ_{inter} , and introduction rate $\eta_T = N_T/\Phi$ of the electronic levels observed in 4H-SiC irradiated with a proton fluence of $3.2 \times 10^{13} \text{ cm}^{-2}$.

Trap label	E_T (eV)	$N_T(\text{cm}^{-3})$	$\sigma_{\text{inter}}(\text{cm}^2)$	$\eta_T(\text{cm}^{-1})$	Comparison with published data	
					Deep level	Attribution
S0	0.18	1.0×10^{13}	2×10^{-14}	~ 0	Ti, ^a P_1/P_2 ^b	Ti impurity ^{a,c} , primary defects ^b
S1	0.20	6.3×10^{13}	7×10^{-18}	3.2	...	...
S2	0.40	1.3×10^{14}	1×10^{-15}	4.1	EH1 ^d	...
S3	0.72	3.5×10^{14}	2×10^{-14}	11.0	Z_1/Z_2 , ^{a,b} EH2 ^d	$N+I_C$ ^c
S4	0.76	1.1×10^{14}	1×10^{-14}	3.5	EH3 ^d	...
S5	1.09	7.7×10^{13}	5×10^{-13}	2.4	RD1/2, ^a EH5 ^d	V_C+V_{Si} ^b

^aReference 7.

^bReference 21.

^cPresent work.

^dReference 13.

^eReference 8.

Charge collection efficiency (CCE) was monitored in measurements which are dealt with more in depth elsewhere,⁵ and it showed that the defects introduced by irradiation yield a significant reduction of transport by diffusion, whereas transport by drift is not affected even at the highest investigated fluences.

As far as the samples irradiated with protons are concerned, a deep-level S0, with enthalpy $E_T=0.18$ eV and concentration equal to $\sim 10^{13}$ cm⁻³, was detected in the as-prepared sample. Upon proton irradiation further levels appeared, i.e., S1 with $E_T=0.20$ eV, S2 with $E_T=0.40$ eV, S3 with $E_T=0.72$ eV, S4 with $E_T=0.76$ eV, and S5 with $E_T=1.09$ eV, the concentration of which was found to increase linearly with the irradiation fluence Φ , as evidenced in Fig. 2. The density of level S0 remained constant at $N_T \approx 10^{13}$ cm⁻³ whatever the fluence was. The parameters of the deep levels and the comparison with data from the literature are given in Table II, while the evolution of the spectra with respect to the irradiation dose is shown in Fig. 3(a). In the diodes irradiated with the highest fluence, 3.2×10^{13} cm⁻², the total deep-level density results are comparable to the doping density, though somewhat lower, so that compensation effects should be expected. These effects were confirmed by subsequent *C-V* measurements, as reported below.

In diodes irradiated by electrons, the deep-level generation occurs with characteristics quite similar to proton irradiation [Fig. 3(b)]. Also in these epilayers, a shallower-level S0 (enthalpy $E_T=0.15$ eV) is found in the as-prepared material. Slightly different from the previous case, we were able to detect also a deeper level, S5*, with $E_T=0.89$ eV. Upon irradiation DLTS detected levels S1* with $E_T=0.23$ eV, S2 with $E_T=0.39$ eV, S3 with $E_T=0.5/0.65$ eV, and S4 with $E_T=0.76$ eV. The parameters of the relevant deep levels are reported in Table III. It is worth noting that some of the deep levels increasing dramatically in concentration in the irradiated diodes are present also before irradiation (continuous lines in Fig. 3) even though with density close to the resolution limit of the experimental setup. The concentration of the levels S1*–S5* is linearly dependent on the irradiation dose, whereas the concentration of S0 is independent of it.

Regarding the electron-irradiated diodes, a low-temperature annealing was observed, the interpretation of which is dealt with in a previous work.⁶ The annealing process consists of two stages. In the first stage, from 360 to 400 K, the amplitude of a DLTS peak labeled S2A (Table III) decreased dramatically while that of peak S3 slightly increased. In the temperature interval where S2A was detected, a minor peak, labeled S2, with equal enthalpy appeared. In the second stage, from 400 to 470 K, the shape of peak S3 sharpened, its amplitude slightly decreased and its enthalpy changed from 0.5 to 0.65 eV, taking on the parameters of the peak Z₁/Z₂ known from the literature.^{7,8} In Fig. 3 only the spectra recorded after annealing are shown, since in the present work we consider important the deep-level parameters found after both annealing stages, just as the analyzed proton-irradiated diodes had been previously annealed before the DLTS analysis.

From the aforementioned results, we can assess that irradiation with either particle type generates almost the same

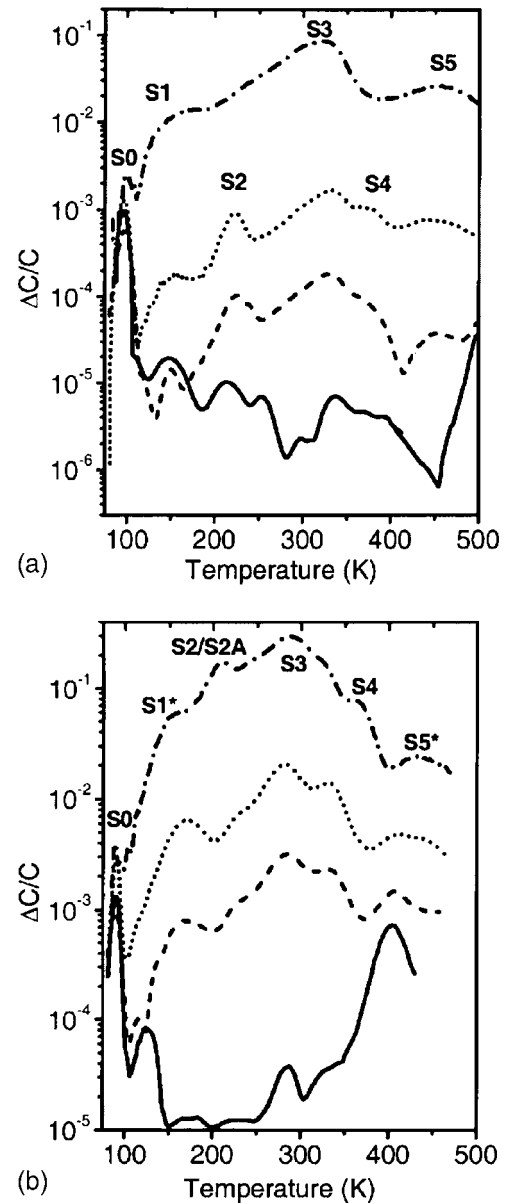


FIG. 3. Evolution of the DLTS spectra. (a) Proton-irradiated diodes. Continuous line: as-prepared diode; dashed line: diode irradiated with $\Phi = 10^{11}$ cm⁻²; dotted line: diode irradiated with $\Phi = 10^{12}$ cm⁻²; and dashed-dotted line: diode irradiated with $\Phi = 3.2 \times 10^{13}$ cm⁻². The emission rate is $e_n = 232$ s⁻¹. (b) Electron-irradiated diodes. Continuous line: as-prepared diode; dashed line: diode irradiated with $\Phi = 4.75 \times 10^{13}$ cm⁻²; dotted line: diode irradiated with $\Phi = 2.37 \times 10^{14}$ cm⁻²; and dashed-dotted line: diode irradiated with $\Phi = 9.50 \times 10^{14}$ cm⁻². The emission rate is $e_n = 46.5$ s⁻¹.

set of electron traps in *n*-type epilayers, as is also stressed by the Arrhenius plot (Fig. 4). The introduction rate $\eta_T = N_T/\Phi$ is shown in Fig. 5 for levels with similar signatures. It is to be noted that η_T is about one order of magnitude higher in the case of proton irradiation, i.e., $\eta_{T,\text{proton}}/\eta_{T,\text{electron}} \sim 10$ for levels S2, S3, and S4, while in the case of levels S1/S1* and S5/S5* $\eta_{T,\text{proton}}/\eta_{T,\text{electron}} \geq 30$. This strongly suggests that (i) defect generation occurs mostly according to the same mechanism in both cases since the same defects are generated and (ii) most of the deep levels detected by DLTS should be either of intrinsic nature or very simple defect complexes involving shallow donor impurities. The former

TABLE III. Enthalpy E_T , concentration N_T , apparent capture cross section σ_{inter} , and introduction rate $\eta_T = N_T/\Phi$ of the electronic levels observed in 4H-SiC irradiated with an electron fluence of $9.5 \times 10^{14} \text{ cm}^{-2}$.

Trap label	E_T (eV)	$N_T(\text{cm}^{-3})$	$\sigma_{\text{inter}}(\text{cm}^2)$	$\eta_T(\text{cm}^{-1})$	Comparison with published data	
					Deep level	Attribution
S0	0.15	1.4×10^{13}	6×10^{-16}	~ 0	Ti, ^a P_1/P_2 ^b	Ti impurity, ^{a,c} primary defects ^b
S1*	0.23	5.5×10^{13}	9×10^{-16}	0.06	...	...
S2A	0.33	6.9×10^{14}	7×10^{-16}	0.73	...	I_C ^d
S2	0.39	4.0×10^{14}	2×10^{-15}	0.42	EH1 ^e	...
S3	0.65/0.50	$4.2/5.4 \times 10^{14}$	$10^{-16}/10^{-14}$	0.44	Z_1/Z_2 , ^{a,b} EH2 ^e	$N+I_C$ ^f
S4	0.75	2.6×10^{14}	6×10^{-15}	0.27	EH3 ^e	...
S5*	0.89	4.6×10^{13}	7×10^{-15}	0.09	RD1/2, ^a EH5 ^e	V_C+V_{Si} ^b

^aReference 7.

^bReference 21.

^cPresent work.

^dReference 6.

^eReference 13.

^fReference 8.

result is not obvious, given the significant difference in the impinging particle masses. The latter result comes from the observation that the more complex the defect configuration is, the more different should be the introduction kinetics in either impinging particle type. From Fig. 5 we also observe that the introduction rate is the smallest for both levels S1*/S1 and S5*/S5. This is also the case in which the ratio $\eta_{T,\text{proton}}/\eta_{T,\text{electron}}$ is the highest. Although the identification of the defects giving rise to deep levels is in many cases still dubious, some can be advanced. Some levels have been extensively studied in the past few years due to their particular annealing behavior or to other characteristics, such as the negative- U behavior,⁹ that made them particularly suitable to a more in-depth analysis. We list in the following our findings on the detected levels (Figs. 3 and 4).

A. Level S0

The behavior of S0 ($E_T=0.15/0.18$ eV), which is the only level maintaining a constant concentration throughout the whole range of fluences, suggests a relationship of this level with impurity atoms, which are not affected at all by irradiation. The literature data indicate that S0 is to be put in

a relationship with an acceptor level originated by the Ti impurity, for which two levels with enthalpy $E_T=0.12$ eV and $E_T=0.16$ eV are attested.^{4,7,10} The capture cross section measured in the proton-irradiated diode ($\sigma=2 \times 10^{-14} \text{ cm}^2$) is significantly larger than that measured in the electron-irradiated diode ($\sigma=6 \times 10^{-16} \text{ cm}^2$). This is probably due to the large uncertainty associated with the determination of the capture cross section by extrapolation to $T=\infty$ of the Arrhenius plot.¹¹ This may also explain the difference between our values and that found by Dalibor *et al.* ($\sigma=1 \times 10^{-16} \text{ cm}^2$).⁷ Relying more on the enthalpy values, we are therefore confident that S0 has to be associated with the Ti impurity. Indeed, Ti may unintentionally contaminate SiC during the growth due to the crucible graphite parts involved.¹²

B. Levels S1 and S1*

In both cases of proton and electron irradiations a deep level is found in the temperature region $T \sim 150$ K. However, some facts suggest that this level is not to be identified with the same defect in one or the other case. First, the deep-level parameters are different enough (Tables II and III) to yield two well distinguishable signatures on the Arrhenius plot. Second, the ratio of the introduction rates for proton and electron irradiations here assumes the value of about 60, thus deviating significantly from the value of about 10 found for

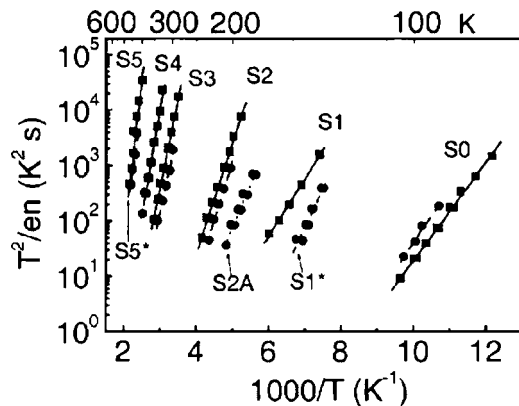


FIG. 4. Arrhenius plot of proton- and electron-irradiated samples, showing that similar sets of traps are present for irradiation with either particle type. The squares refer to the proton irradiation and the circles to the electron irradiation. The solid lines represent the fit for the proton irradiation and the dashed lines the fit for the electron irradiation.

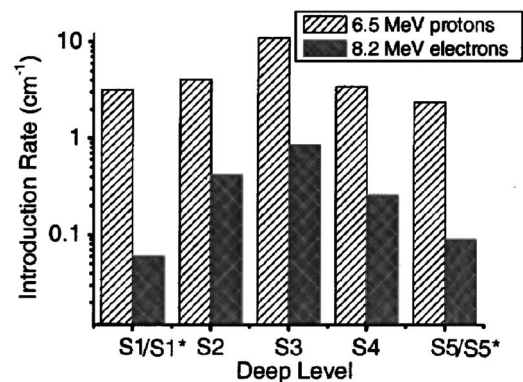


FIG. 5. Introduction rate η_T of deep levels introduced by proton (light columns) and electron (dark columns) irradiations.

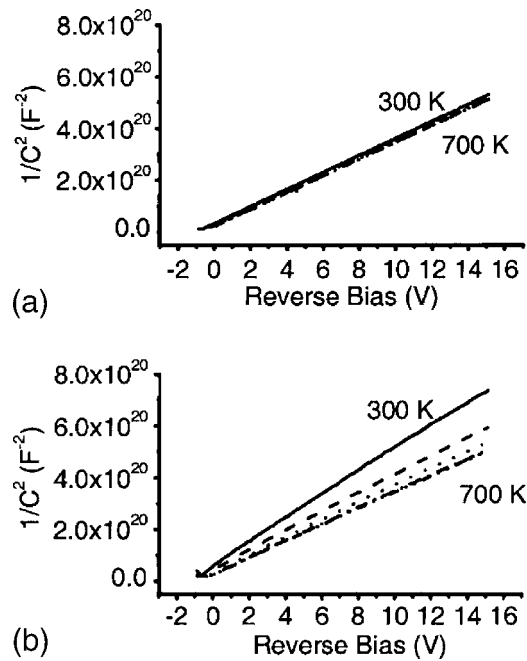


FIG. 6. C^2 - V plots of (a) as-prepared sample and (b) diode irradiated with $3.2 \times 10^{13} \text{ cm}^{-2}$ 6.5-MeV protons, showing free-charge-carrier compensation. The characteristics measured from 300 to 700 K are shown, with 100-K step.

all levels with the same deep-level parameters except for S5. Thus, we conclude that S1 and S1* should not be identified with the same level, i.e., electrons and protons introduce two distinguished complex defect configurations with close enthalpy.

C. Levels S2 and S2A

The peculiarity of level S2A ($E_T=0.33$ eV) is its low temperature annealing out.⁶ The amplitude of the peak related to this level, in fact, decreases dramatically in the DLTS spectrum after the sample enters the temperature intervals of 360–400 K. Annealing out of a level lying close in enthalpy, labeled EH1 in the literature, was previously reported in 2.5-MeV electron-irradiated samples, although at a higher temperature (750 °C).¹³ Level S2A may be related to the carbon interstitial I_C .^{6,14,15} After the annealing, level S2 is still present in the DLTS spectrum of electron-irradiated diodes, presenting an enthalpy close to S2A and a higher capture cross section (Fig. 4). This level matches up very well with level S2 found in the proton-irradiated samples and with the EH1 level.

D. Level S3

It is known that in the region of the DLTS spectrum around 300 K a dominating peak with activation energy $E_T=0.6/0.7$ eV appears on irradiation,^{4,7,9} although it can be found also in as-prepared samples.¹⁶ It is generally believed to be a set of discrete levels with similar energy, some of which are unstable and change configuration after: (i) months at room temperature,¹⁷ (ii) thermal treatments,^{6,9} and (iii) combined action of thermal annealing and reverse bias.¹⁸ The level Z_1/Z_2 , which is the most stable of these levels,

(called S3 in the present work) has been in the last few years the most investigated electronic level both from the experimental and the theoretical point of view. According to recent studies^{8,9,19} Z_1/Z_2 is a negative- U center and is related to a dicarbon interstitial next to a nitrogen atom^{8,19} or, alternatively, to an intrinsic defect complex possibly involving hydrogen.²⁰

E. Level S4

From the analysis of the Arrhenius plot and of the introduction rates, it is apparent that also level S4 is associated with the same defect in both cases of proton and electron irradiations. This level peak, significantly overlapping part of the dominating peak associated with level S3, should also be associated with an intrinsic defect or to a small complex.

F. Levels S5 and S5*

These levels are part of a group of levels denominated in the literature as “radiation damages” (RD),^{7,21} so as to stress their irradiation-induced origin. This was confirmed by our observations, as we found the concentration of both levels S5 ($E_T=1.09$ eV) and S5* ($E_T=0.89$ eV) to grow linearly with the irradiation dose. However, we observed that level S5* is sometimes present in significant amount also in as-prepared samples. The microscopic structure of these defects has been studied in previous works.²¹ According to those authors, RD1/2 ($E_T=0.96$ eV), quite close to the levels here denominated S5 and S5*, is related to a defect complex involving a carbon and a silicon vacancy V_C+V_{Si} . The attribution to a small defect complex is in agreement with the fact that this level can be found also in nonirradiated samples. The ratio of the introduction rates between proton- and electron-irradiated samples is about 30, much higher than the value of 10 found for the pairs of levels matching up together in the Arrhenius analysis. From this it turns out that S5 and S5* could be associated with different defects or to the same complex defect presenting very different generation kinetics depending on the impinging particle type.

The compensation of free charge carriers due to the presence in the gap of defect-related deep levels has been monitored in both cases of proton and electron irradiations by means of C - V characteristics. Significant compensation effects were observed mainly in the samples irradiated with the highest doses, i.e., $3.2 \times 10^{13} \text{ cm}^{-2}$ for proton irradiation and $9.5 \times 10^{14} \text{ cm}^{-2}$ for electron irradiation. In the proton-irradiated diodes, C - V measurements (Fig. 6) were carried out in the temperature intervals of 300–700 K. Both in the as-prepared samples and in the samples irradiated with fluences up to 10^{12} cm^{-2} , the free-charge-carrier concentration remained constant at a value close to $n=4.5 \times 10^{15} \text{ cm}^{-3}$ throughout the whole temperature interval. On the contrary, in the sample irradiated with a $3.2 \times 10^{13} \text{ cm}^{-2}$ proton dose the free-charge-carrier concentration varied from $n=3.3 \times 10^{15} \text{ cm}^{-3}$ at $T=300$ K to $n=4.7 \times 10^{15} \text{ cm}^{-3}$ at $T=700$ K. The original free charge density is restored at $T \sim 700$ K even in highly compensated samples. This result disagrees with the compensation model,²² according to which free charge compensation should be mainly due to nitrogen pas-

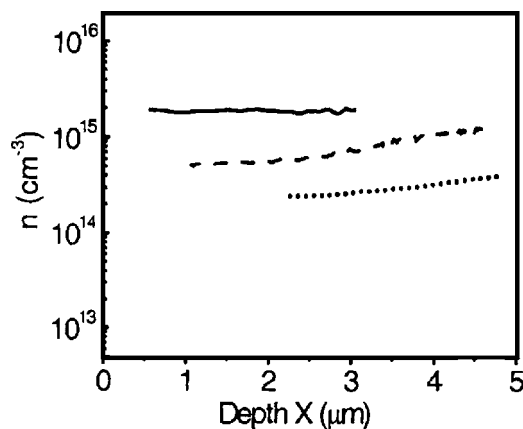


FIG. 7. Free-carrier density compensation and recovery in electron-irradiated diodes. Continuous line: as-prepared diode; dotted line: diode irradiated with $9.5 \times 10^{14} \text{ cm}^{-2}$, before annealing at $T=360\text{--}400 \text{ K}$; and dashed line: the same diode after annealing at $T=360\text{--}400 \text{ K}$.

sivation by defect complexes induced by irradiation. The N atoms are therefore subtracted from their role of active shallow doping centers. If such defects were reasonably stable, they would not dissociate for a relatively low temperature such as $T=700 \text{ K}$, and there should not be the possibility of a recovery of free charge density to the same value observed before irradiation.

Figure 7 shows the link between the annealing steps observed by DLTS and the net free-carrier recovery. Upon irradiation we observed strong compensation effects, as the free-carrier density, calculated by the standard procedure,²³ decreased from $n=2.0 \times 10^{15} \text{ cm}^{-3}$ at a distance from the Schottky junction $x_d=2 \mu\text{m}$ to $n=2.5 \times 10^{14} \text{ cm}^{-3}$ at the same distance. After the annealing stage between 360 and 400 K, the free-carrier density increased to the value $n \sim 5 \times 10^{14} \text{ cm}^{-3}$, which is consistent with the amplitude reduction of peak S2A/S2 in the DLTS spectrum and in agreement with the interpretation of the annealing out of level S2A as an annihilation of simple point defects.⁶

IV. CONCLUSIONS

Irradiation with either protons or electrons, at the energies and fluences examined in this work, introduces in the band gap of 4H-SiC several deep levels, producing free-charge-carrier compensation, which is dramatic for the highest fluences used. Nevertheless, the material confirms to be highly radiation hard, as its transport characteristics give evidence: irradiation-induced defects tend indeed to limit the charge transport in diffusion regime, but do not have an equally significant influence on the transport in drift regime. DLTS shows that a few levels (S0 and S5) are already present in detectable concentrations ($\sim 10^{13} \text{ cm}^{-3}$) in the as-prepared diodes. By irradiation another set of levels is either introduced or enhanced in concentration (S1–S5), growing with the irradiation dose. The only level maintaining a constant concentration is S0, likely related to impurity Ti atoms rather than to primary defects. Most deep levels generated by both particles are the same, as deduced by the Arrhenius and introduction rate analyses. Generally speaking, protons tend to introduce a density of defects about one order of magni-

tude (or more) higher than electrons of similar energy, as in agreement with the experimental and theoretical results comparing the nonionizing energy loss (NIEL) of either particle type²⁴ in SiC. On the one hand, one may expect this change in the introduction rates, considering the different particle masses; on the other hand, it is not obvious that such a mass and charge difference yields the same deep levels. Thus, the introduced levels must be elementary intrinsic defects or complexes involving single N or H atoms. Other levels measured for different particle irradiation such as S1 compared with S1* ($E_T \sim 0.2 \text{ eV}$) and S5 compared with S5* ($E_T \sim 0.9\text{--}1.1 \text{ eV}$) are unlikely ascribable to the same kind of defects, unless to consider an alternative generation kinetics depending upon the irradiation source.

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